

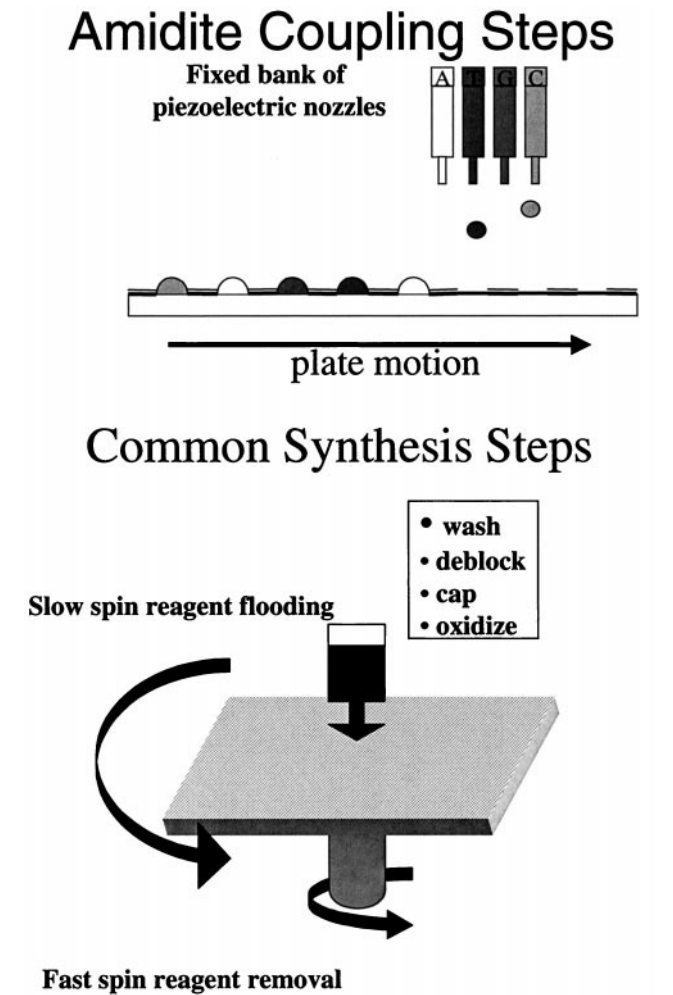


FIG. 3. Activated phosphoramidite monomers are delivered to the array features by piezoelectric nozzles under the control of a command file that guides synthesis of the correct sequence at each array location.

Washing, deblocking, capping, and oxidizing reagents are flooded onto the surface and centrifuged off between steps.

hybridization arrays or microfluidic arrays, are sure to play a significant role.

In a demonstration project, the polymorphic human *NAT2* gene, a biomarker for cancer and drug metabolism, was selected as a model system to develop general strategies for designing and optimizing DNA microarrays. The goal was to develop homozygous and het-

erozygous DNA genotyping tests to yield a maximum of high-confidence genotype assignments. The approaches developed using this model can be generalized to design hybridization arrays of virtually any complexity for genotyping and other applications, including gene expression profiling. Furthermore, once an array is designed, it can be expanded in scope and complexity without sacrificing the development invested in the original array.

Materials and Methods

The scope of the genotyping assay design process includes the following steps: 1) selecting the polymorphisms to be genotyped; 2) collecting the sequence contexts around each polymorphism; 3) selecting the target labeling and hybridization conditions; 4) designing the hybridization probes that discriminate among the possible genotypes; 5) testing the assay performance with samples of known genotype; and 6) redesigning the probe set until the maximum genotyping performance is obtained under the chosen conditions. Once this process is complete, final test validation requires that an unknown sample set be genotyped using the microarray assay and the results to be confirmed using a reference genotyping method such as direct Sanger sequencing.

Surface tension DNA array synthesis is the combination of two processes, substrate surface preparation and in situ DNA synthesis. The substrate preparation begins with glass cleaning followed by spin coating with a layer of photoresist. Photolithographic patterning with a mask to define the desired size and distribution of the array features follows spin coating. This process is illustrated in Fig. 1.

The patterned arrays are developed then immersed in a solution of fluorosilane to generate a hydrophobic silane layer surrounding the array features, which are still protected with photoresist. After the fluorosilane is cured, acetone treatment removes the remaining photoresist exposing the array feature sites. The features are coated with an aminosilane then coupled with a linker molecule that will support subsequent DNA oligonucleotide synthesis. These processed substrates display the surface tension behavior for aqueous and polar organic solvents shown in Fig. 2.

The surface tension patterned substrates are aligned on a chuck in the robotic array synthesizer where piezoelectric nozzles are used to deliver solutions of activated standard DNA synthesis amidites as shown in Fig. 3. Washing, deblocking, capping, and oxidizing reagents are delivered by bulk flooding the reagent onto the substrate surface and spinning the chuck mount to remove excess reagents between reactions. The substrate surface is environmentally protected throughout the synthesis by a blanket of dry N₂ gas. Localizing and metering amidite delivery is mediated by a computer command file that directs delivery of the four amidites during each pass of the piezoelectric nozzle bank so the predetermined oligonucleotide is synthesized at each array coordinate. Array design iterations are accomplished by altering this synthesis command file.

DNA array probe design was aimed at discriminating the seven polymorphic sites in the human *N*-acetyltransferase gene shown in Table 1 (*NAT2*,

TABLE 1
The seven common polymorphisms in the human *NAT2* gene and the combinations that define each of the most common alleles

Allele	Polymorphism							Phenotype	Frequency
	G191A	C282T	T341C	C481T	G590A	A803G	G857A		
*4	G	C	T	C	G	A	G	Rapid	23.40
*5A	G	C	C	T	G	A	G	Slow	2.50
*5B	G	C	C	T	G	G	G	Slow	40.90
*5C	G	C	C	C	G	G	G	Slow	2.60
*6A	G	T	T	C	A	A	G	Slow	28.40
*6B	G	C	T	C	A	A	G	Slow	Rare
*7A	G	C	T	C	G	A	A	Slow	Rare
*7B	G	T	T	C	G	A	A	Slow	2.10
*12A	G	C	T	C	G	G	G	Rapid	0.10
*12B	G	C	T	C	G	A	G	Rapid	Rare
*13	G	T	T	C	G	A	G	Rapid	Rare
*14A	A	C	T	C	G	A	G	Slow	Rare
*14B	A	T	T	C	G	A	G	Slow	0.10